

Grover's Search and Quantum Counting

Project 6 - PHYS-541 Quantum Computing

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Project Overview & Objectives

This project focuses on the implementation and analysis of Grover's Search and Quantum Counting algorithms.

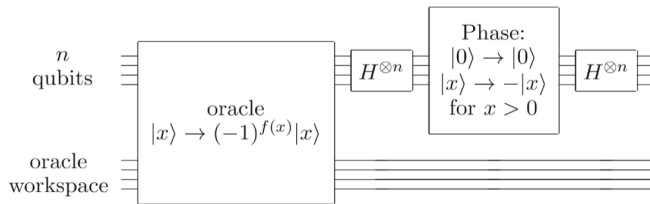
- ➊ **Theory:** Detailed explanation of Grover's algorithm and Quantum Counting (based on Nielsen & Chuang).
- ➋ **Complexity:** Analysis of quantum speedup limits and NP complexity class.
- ➌ **Implementation:** Construction of Oracles (with and without Ancilla) on IBM Qiskit.
- ➍ **Noise Analysis:** Performance study under simulated noise and circuit depth limitations.

1. Grover's Algorithm: The Mechanism

Goal: Find marked element(s) x such that $f(x) = 1$ in an unsorted database of size $N = 2^n$. (ex : Sudoku, Traveling Salesman Problem)

The algorithm iterates the **Grover Operator** $\hat{G} = (2|\psi\rangle\langle\psi| - I)\hat{O}$, where $|\psi\rangle = H^{\otimes n}|0\rangle$:

- **Oracle ($\hat{O}$):** Marks the solution by flipping its phase: $\hat{O}|x\rangle = (-1)^{f(x)}|x\rangle$
- **Diffuser:** Performs an inversion about the mean amplitude, amplifying the probability of the marked state.



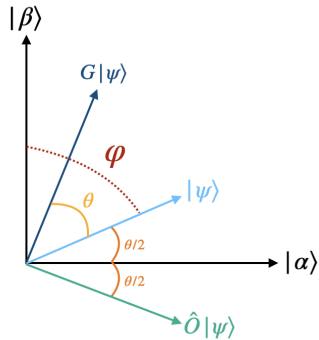
L. K. Grover, Phys. Rev. Lett. 79, 325 (1997).

M. A. Nielsen & I. L. Chuang, *Quantum Computation and Quantum Information* (2010).

1. Grover's Algorithm: Geometric Picture

Key Idea: Grover's algorithm acts as a **rotation** in a 2D subspace, progressively amplifying the target state.

- The quantum state lives in the plane spanned by:
 - $|\beta\rangle$: superposition of **solutions**
 - $|\alpha\rangle$: superposition of **non-solutions**
- Each Grover iteration rotates the state by an angle θ
- After an optimal number of iterations, the state aligns with $|\beta\rangle$
- Optimal Number of Iterations: $r_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{m}}$



Rotation in the subspace spanned by $|\alpha\rangle$ and $|\beta\rangle$.

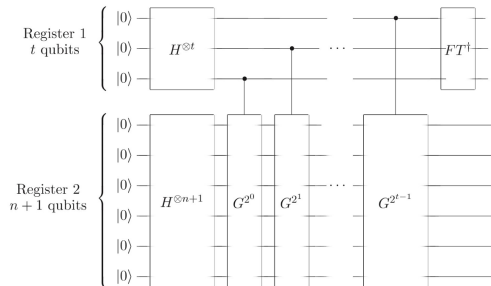
1. Quantum Counting Algorithm

Problem: Grover's optimal iterations r depend on m . What if m is unknown?

The Solution

Combine Grover operator G with **Quantum Phase Estimation (QPE)**.

- G has eigenvalues $e^{\pm i\theta}$.
- **Phase Kickback** : transfer θ into the counting register.
- **Inverse QFT** : reveals the phase θ .
- we measure m : $m \approx N \sin^2(\theta/2)$



2. Computational Complexity : Limits of Speedup

1. The Quadratic Speedup

- Classical: $\mathcal{O}(N)$
- Grover: $\mathcal{O}(\sqrt{N})$

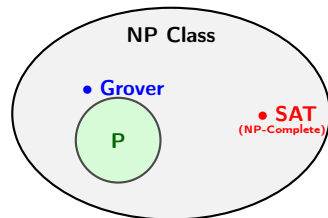
2. Is further speedup possible?

BBBV Theorem (1997)

Proven lower bound: Any quantum black-box search requires $\Omega(\sqrt{N})$ queries.

Consequence: Grover is **asymptotically optimal**.
We cannot do better than $\sqrt{N}$.

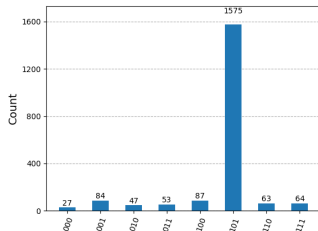
3. Can it solve NP-Complete?



- Brute Force: $\mathcal{O}(2^n)$
- Grover: $\mathcal{O}(2^{n/2})$
 - Still **Exponential**. Grover does *not* solve NP-Complete efficiently.

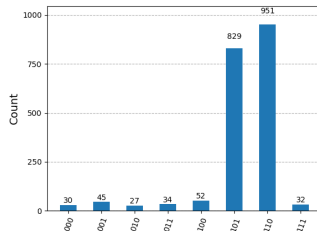
3. Implementation of Grover's Algorithm

Case A: 1 Solution ($|101\rangle$)



Clear peaks at phases corresponding to $m = 1$ (ibm_torino (133 qubits)).

Case B: 2 Solutions ($|101\rangle, |110\rangle$)



Peaks shift location, indicating a larger angle θ (ibm_torino (133 qubits)).

Protocol

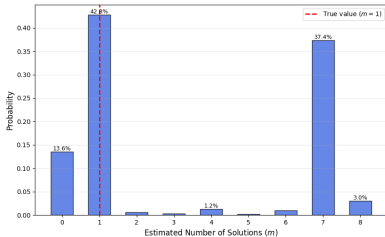
- **System:** $n = 3$ qubits ($N = 8$).
- **Shots:** 2000 executions per circuit.

Key Finding: Distinct Signatures

Despite hardware noise (decoherence), the algorithm successfully distinguishes the solution counts:

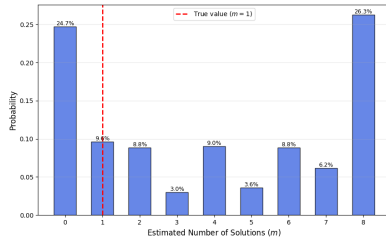
3. QPE: The Depth Limit

Ideal Simulation



Well-resolved phase peaks ($n = 3$, $t = 6$, AerSimulator).

Real Hardware



Signal completely washed out ($n = 3$, $t = 6$, ibm_torino).

Why does QPE fail on NISQ hardware?

- QPE applies controlled powers of Grover: G^{2^k}
- Maximum circuit depth scales as:
$$\text{Depth} \sim 2^{t_{\text{bits}}}$$
- Exponential depth $\Rightarrow$ coherence time T_2 is exceeded $\Rightarrow$ Quantum Counting is not NISQ-ready

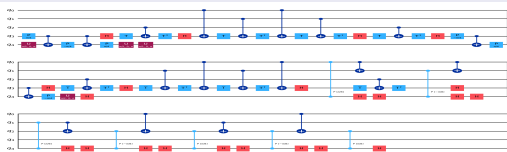
3. Oracle Implementation Strategy

Task: Construct the marking function $f(x)$ for specific targets (e.g., $|111\rangle$).

A. No Ancilla (MCXGrayCode)

Decomposes using relative phase and recursion.

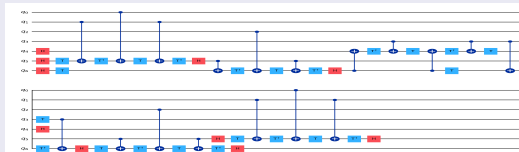
- **Pros:** Minimizes qubit count (good for simulators).
- **Cons:** Quadratic Depth $\mathcal{O}(n^2)$.



B. With Ancilla (MCXVChain)

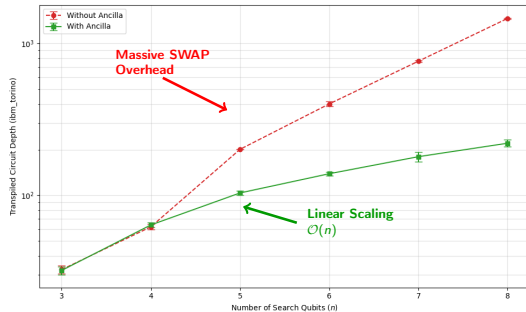
Uses helper qubits (V-Chain) to break connections.

- **Pros:** Linear Depth $\mathcal{O}(n)$.
- **Cons:** Requires extra qubits.



3. Hardware Complexity: The Cost of Connectivity

Setup: Backend `ibm_torino` (133-qubit) | Metric: Transpiled Depth (Log Scale)



Key Finding: The Connectivity Bottleneck

Without ancillas, the transpiler must insert massive SWAP chains to link distant qubits. This overhead causes the "No Ancilla" approach to hit the T_2 coherence limit as soon as $n > 5$.

4. Hardware Performance: Grover Oscillations

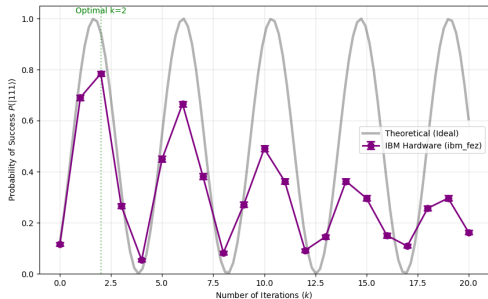


Fig : ($n = 3$, target = $|111\rangle$, shots = 2 000, ibm_torino (133 qubits))

Key Observations:

- 1 **Validation:** The hardware (purple) closely follows the theoretical sinusoidal curve (black).
- 2 **Optimal Peak:** Maximum probability occurs at $k = 2$, matching the theory exactly.
- 3 **Damping:** The peak < 1.0 and decays over time. As k increases, circuit depth increases, accumulating gate errors and decoherence.

4. Performance Under Noise

The Challenge: Small $\theta/2 \approx \sqrt{m/N}$ implies large number of iterations r .

$$\text{Circuit Depth} \propto r \times \text{Oracle Depth}$$

Simulation Results:

- For small n ($n \leq 4$): High fidelity.
- For larger n : Success probability drops below classical threshold.
- **Limit:** On current NISQ models, coherence time limits usage to small instances.

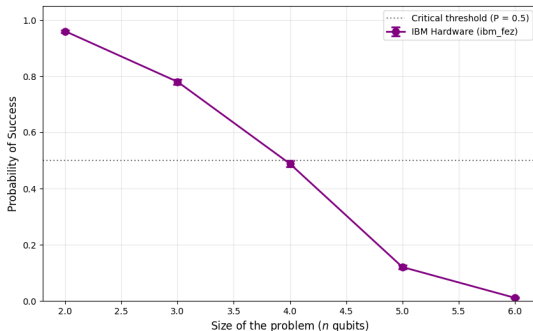


Fig.: Success probability of Grover's algorithm versus qubit number n on `ibm_fez`, at optimal iterations without ancilla ($m = 1$, target $|11 \dots 11\rangle$).

4. Noise Impact: Stability Range Analysis

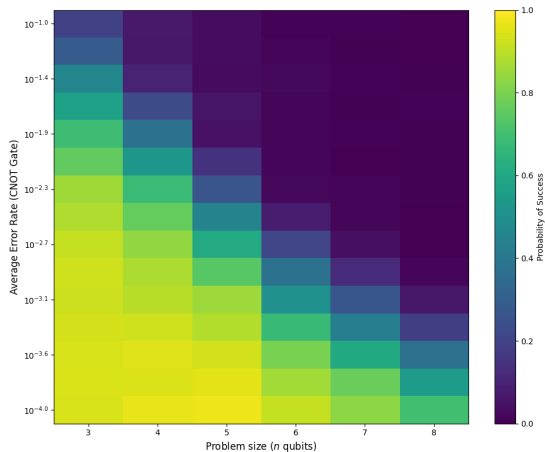


Fig.: Heatmap of the phase transition in Grover's success probability versus qubit number n and CNOT error rate, at optimal iterations without ancilla ($m = 1$, target $|11 \cdots 11\rangle$, AerSimulator).

Conclusion — What We Learn from Grover on NISQ

Oracle Design

Ancilla Required

$$n \geq 4 \ (m = 1)$$

MCXVChain ✓

No-Ancilla ✗

Theory

$\sqrt{N}$ Speedup

Optimal, but still exponential

NP-Complete $\notin$ BQP

Hardware

Noise Wall

$$n \approx 6 \ (m = 1)$$

$$\text{Depth} > T_2$$

Grover is optimal — but not scalable on NISQ hardware

Error correction is required for meaningful scaling

Annexes

A.1. Grover: Theory and Optimal Performance

Subspace Definitions: For m solutions in a database of size $N = 2^n$:

$$|\beta\rangle = \frac{1}{\sqrt{m}} \sum_{x \in S} |x\rangle \quad |\alpha\rangle = \frac{1}{\sqrt{N-m}} \sum_{x \notin S} |x\rangle$$

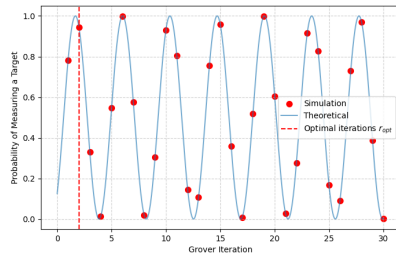
Initial State Decomposition: $|\psi\rangle = \cos(\theta/2)|\alpha\rangle + \sin(\theta/2)|\beta\rangle$, $\theta = 2 \arcsin\left(\sqrt{\frac{m}{N}}\right)$

Success Probability: $P(r) = \sin^2\left((2r+1)\frac{\theta}{2}\right)$ **Optimal Number of Iterations:** $r_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{m}}$

Example: $n = 3$, $m = 1$

• $r_{\text{opt}} \approx 2$

• $P_{\text{success}} \approx 0.945$



Theoretical success probability vs Grover iterations.

A.2. Quantum Phase Estimation (QPE) : Ideal vs Noisy Simulation

Simulation Setup

- **System:** $n = 3$, $m = 1$ solutions.
- **Precision:** $t = 6$ qubits.
- **Noise:** Depolarizing ($\epsilon = 10^{-5}$).

Spectral Interpretation: The Grover operator G acts as a rotation in a 2D subspace, possessing conjugate eigenvalues $e^{\pm i\theta}$. QPE projects the state onto both eigenphases:

- Peak 1: $\phi_1 \approx \theta \rightarrow \text{integer } j$
- Peak 2: $\phi_2 \approx 2\pi - \theta \rightarrow \text{integer } 2^t - j$

Solution Invariance: Due to the symmetry $\sin^2(x) = \sin^2(\pi - x)$, both peaks yield the exact same count:

$$m = N \sin^2\left(\frac{\theta}{2}\right) = N \sin^2\left(\frac{2\pi - \theta}{2}\right)$$

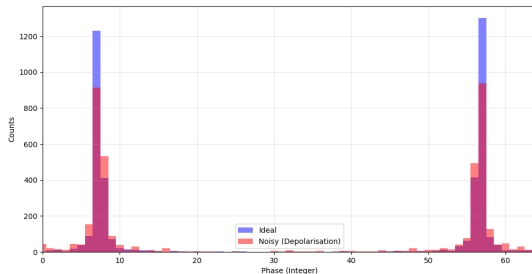


Fig: Number of targets (m). The two symmetric peaks physically correspond to the forward and backward rotation of the Grover operator. ($n = 3, j = \{6, 58\}$, shots = 2 000, AerSimulator())

A.3. Grover vs. Shor Strategy

Grover's Algorithm

The "Brute Force" Booster

- **Problem:** Unstructured Search (Database, SAT).
- **Structure:** None (Black Box).
- **Speedup:** **Quadratic** $\mathcal{O}(\sqrt{N})$.
- **Impact:** Optimizes search but does **not** break NP-Complete barriers.

Shor's Algorithm

The "Code Breaker"

- **Problem:** Factorization (RSA), Period Finding.
- **Structure:** Exploits **Periodicity**
 $f(x) = f(x + r)$.
- **Speedup:** **Exponential** $\mathcal{O}(\text{poly}(\log N))$.
- **Impact:** Solves Factorization efficiently (In BQP).

Key Difference: *"Grover speeds up the checking process. Shor cheats by finding the hidden mathematical structure."*

A.4. From Grover to Digital Quantum Simulation

1. My Project (Quantum Counting)

Goal: Find the number of solutions m .

- **Operator:** Grover Iterate G .
- **Eigenvalues:** Encoded in phase θ .
- **Relation:** $G|\psi\rangle = e^{\pm i\theta}|\psi\rangle$

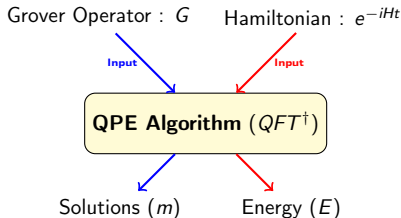
$$m \approx N \sin^2(\theta/2)$$

2. Digital Quantum Simulation

Goal: Find molecular energy E .

- **Operator:** Time Evolution $U = e^{-iHt}$.
- **Eigenvalues:** Encoded in energy.
- **Relation:** $U|\psi\rangle = e^{-iEt}|\psi\rangle$

$$\text{Phase } \phi \propto \text{Energy } E$$



A.5. Future Work: Scaling Grover with QEC

1. The Current Obstacle (NISQ)

Observation: My project failed at $n > 6$.

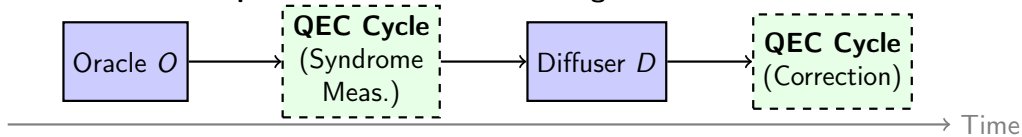
- **Reason:** Grover depth scales as $\mathcal{O}(\sqrt{2^n})$.
- **Physics:** Circuit duration $> T_2$ (Coherence time).
- **Result:** Iterations destroy the quantum state before amplification finishes.

2. The Solution (Fault Tolerance)

Goal: Make "Effective T_2 " infinite.

- **Logical Qubits:** Encode 1 logical qubit into k physical qubits (e.g., Surface Code).
- **Active Correction:** Insert **Stabilizer Checks** between Grover iterations.

Proposed Architecture: The "Logical" Grover Circuit



Stabilizers remove entropy generated during Oracle/Diffuser steps

A.6. The NISQ Alternatives (VQE & QAOA)

Hybrid Algorithms: Pairing Classical Optimization with Quantum Evaluation

The Mechanism (Hybrid Loop)

Quantum Processor acts as a co-processor (GPU).

- 1 **Classical CPU:** Sends parameters θ .
- 2 **Quantum QPU:** Prepares state $|\psi(\theta)\rangle$ and measures Cost C .
- 3 **Classical CPU:** Updates θ to minimize C (Gradient Descent).

Advantage: Short depth, noise resilience.

1. VQE (Chemistry)

Variational Quantum Eigensolver

- **Goal:** Find Ground State Energy.
- **Use Case:** Drug discovery, Material science.
- *Alternative to Phase Estimation.*

2. QAOA (Optimization)

Quantum Approx. Optimization Algo

- **Goal:** Solve combinatorial problems (MaxCut, TSP).
- **Structure:** Alternating "Cost" and "Mixer" layers.

1. Grover's Algorithm: Geometric Interpretation

Geometric Interpretation: Rotates the state vector by angle θ towards the target state $|\beta\rangle$.

1. Subspace Definitions: Let S be the set of m solutions. We define the normalized basis states:

$$|\beta\rangle = \frac{1}{\sqrt{m}} \sum_{x \in S} |x\rangle \quad (\text{Target superpos.})$$

$$|\alpha\rangle = \frac{1}{\sqrt{N-m}} \sum_{x \notin S} |x\rangle \quad (\text{Non-target})$$

2. Initial State Decomposition: The initial state $|\psi\rangle = H^{\otimes n}|0\rangle$ lies in the plane spanned by $|\alpha\rangle$ and $|\beta\rangle$:

$$|\psi\rangle = \sqrt{\frac{N-m}{N}} |\alpha\rangle + \sqrt{\frac{m}{N}} |\beta\rangle$$

$$|\psi\rangle = \cos(\theta/2) |\alpha\rangle + \sin(\theta/2) |\beta\rangle$$

3. Geometric Action: The Grover operator G performs a rotation of angle θ in this 2D plane:

$$\theta = 2 \arcsin(\sqrt{m/N}) \approx 2\sqrt{m/N}$$

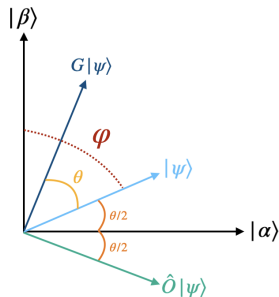


Fig: Geometric interpretation of the rotation in the subspace spanned by $|\alpha\rangle$ and $|\beta\rangle$.

1. Grover's Algorithm: Theoretical Performance

Number of Iterations: To maximize the probability of measuring a marked state, perform:

$$r \approx \frac{\pi}{4} \sqrt{\frac{N}{m}}$$

- **Constraint:** Knowing m is required to stop at the peak.
- **Link to Project:** This motivates **Quantum Counting** to estimate m when unknown, preventing over-rotation.

Probability of Success:

$$P(r) = \sin^2 \left((2r + 1) \frac{\theta}{2} \right)$$

Example: $n = 3$ ($N = 8$), $m = 1$:

- $r_{\text{opt}} \approx 2$ iterations.
- $P_{\text{success}} \approx 0.9453$.

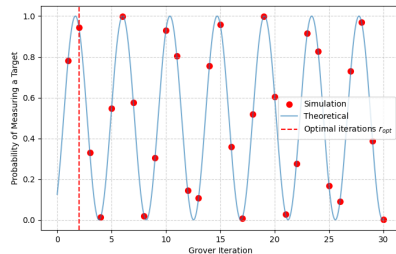


Fig : Probability of measuring the target state $|101\rangle$ vs number of Grover iterations r for $n = 3$, $m = 1$.